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# Divyajyoti

## Education

2019-24 **Ph.D. (Gravitational-wave physics and astronomy)**, *Indian Institute of Technology (IIT) Madras*, Chennai, India
Thesis: Synergies in analysing binary black hole mergers: effect of orbital eccentricity, spin-precession, and non-quadrupole modes. Advisor: Asst. Prof. Chandra Kant Mishra

2015-19 **Bachelor of Technology (Engineering Physics)**, *National Institute of Technology (NIT) Agartala*, Tripura, India
4$^{th}$ (final) year from IIT Madras

## Research Interests

Data analysis of gravitational wave signals from compact binary mergers in current and future detectors

Gravitational Wave population inference, formation channels of compact binaries

Tests of general relativity

## Skills

*Programming* C/Embedded C/C++, Python, Mathematica

*Software packages* LIGO Algorithm Library Suite (`LALSuite`), PyCBC, `bilby`, `PESummary`, `gwbench`, `gwforge`, `GWFish`, `gwpopulation`, HEASoft, CALDB, SAOImage DS9, `git`, `htcondor`, Slurm, LaTeX

*Platforms* Arduino UNO/mega2560, ATmega16/328, Windows, Linux, macOS

*Languages* English (read/write/speak), Hindi (read/write/speak)

## Experience

Oct 2024 - present **Research Associate**, *School of Physical, Chemical, and Environmental Sciences, Cardiff University*, UK

Jan - July 2024 **Post-doctoral Equivalent Fellow**, *IIT Madras*, Chennai, India

2019 - 2024 **Graduate Research Fellow**, *IIT Madras*, Chennai, India
Thesis: Synergies in analysing binary black hole mergers: effect of orbital eccentricity, spin-precession, and non-quadrupole modes. Advisor: Asst. Prof. Chandra Kant Mishra

2018 - 2019 **Undergraduate Research Project**, *IIT Madras*, Chennai, India
Title: Distance and inclination angle measurements for non-optimally oriented Binary Black Holes using Fisher Matrix Formalism, Advisor: Asst. Prof. Chandra Kant Mishra

June 2018 **Summer Internship**, *Indian Institute of Astrophysics*, Bengaluru, India
Title: Timing studies of Active Galactic Nuclei using NuSTAR, Advisor: Prof. C. S. Stalin

Dec 2017 - Jan 2018 **Winter Internship**, *Raman Research Institute*, Bengaluru, India
Title: Arduino based Oscilloscope for Single Atom Lab (Light and Matter Physics), Advisor: Late Prof. Hema Ramachandran

May - June 2016 **Summer Internship**, *National Institute of Technology Agartala*, Tripura, India
Title: Arduino based Vehicle Tracking System, Advisor: V. Ajeeth

## Invited talks and panels

2026 – *Can gravitational waves constrain star-formation rates in the near future?*, Seminar, ICTS-TIFR, IN

– **Panel:** *Mild eccentricity, major consequences: astrophysical mis-inferences, waveform systematics and tests of GR*, GW-Eccentricity Workshop, Corfu, Greece

– *Mapping the star-formation peak with LIGO-A# and next-gen GW detectors*, GRASP, Utrecht University, Netherlands

2025 – *Probing the star formation peak with BBH observations in the next-gen GW detectors*, Johns Hopkins University, USA

– *Interplay of spins and eccentricity in gravitational wave events*, Gravity Seminars, University of Southampton, UK

– *Role of spins in parameter estimation of eccentric binaries*, Eccentricity workshop, IIT Madras, IN

2023 – *Are we ready for gravitational waves from eccentric binaries?*, Young Researchers' Gravity Meeting, CMI, IN

2022 – *Detection and parameter estimation of eccentric binaries in the upcoming GW observing runs*, Kavli Institute for Cosmology, University of Cambridge, UK

– *Observing higher modes of gravitational radiation in future detectors*, 2$^{nd}$ Chennai Symposium on Gravitation and Cosmology (online), IIT Madras, IN

2021 – *Detectability of gravitational higher order modes in 3G era*, Seminar (online), ICTS-TIFR, IN

2020 – *Astrophysics of higher order modes*, 1$^{st}$ Chennai Symposium on Gravitation and Cosmology, IIT Madras, IN

## Select conference presentations

2026 – LVK Collaboration Meeting, IUCAA, IN

– ET Symposium, Aachen, Germany (poster)

– GW:UK meeting, Nottingham, UK (short talk)

2025 – GWPAW, Georgia Tech, USA (poster)

– Amaldi16/GR24, Glasgow, UK (short talk)

– RAS Specialist Meeting, London, UK (short talk)

2024 – GWPAW, University of Birmingham, UK (short talk, poster)

– LVK Collaboration Meeting, Louisiana State University, USA (poster)

2023 – 10$^{th}$ ICGC, IIT Guwahati, IN (poster)

– Amaldi15 (online talk)

2022 – 32$^{nd}$ meeting of IAGRG, IISER Kolkata, IN (short talk)

– Young Astronomers' Meet, ARIES, IN (poster)

– LVK Collaboration Meeting, Cardiff University, UK (poster)

– $40^{th}$ Annual Meeting of the ASI, IIT Roorkee, IN (short talk)

2021 – Amaldi14 (online talk)

2019 – $9^{th}$ ICGC, IISER Mohali, IN (poster)

## Awards and Fellowships

Jan-July 2024 Awarded **Post-doctoral Equivalent Fellowship (PDEF)** at IIT Madras for submitting PhD thesis 6 months before the scheduled submission date.

Jan 2024 Awarded the **Institute Research Award** by IIT Madras, Chennai "in recognition of excellent PhD work".

July 2022 Awarded the **International Travel Support grant** by Science and Engineering Research Board (SERB), Department of Science and Technology, Govt. of India, to present paper at LVK September meeting in Cardiff, UK.

Nov 2018 Winner of All India Science Fiction writing competition conducted by CSIR, IN.

May 2018 Awarded an upgrade (as rank #1 in Dept. of Physics, NIT Agartala) – transfer to IIT Madras for B.Tech. final year followed by Direct PhD offer.

Dec 2013 Winner of All India Science Fiction writing competition conducted by CSIR, IN

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## Leadership positions, mentorship, and teaching

Leadership Positions
- Jan 2026 - present: Postdoc representative in LIGO Scientific Collaboration (LSC) Academic Advisory Committee (LAAC)
- Dec 2024 - present: Co-chair for the Multimessenger Astronomy group contributing towards the Science Traceability Matrix for Cosmic Explorer
- 2017-18: Student internship coordinator, NIT Agartala
- 2015-18: Editor of "Sanskriti" magazine, NIT Agartala

Mentorship
- May 2025: Mentor for the GW Open Data Workshop - 2025
- June-July 2025: Katie Hermanson, summer Intern at Cardiff University
- July-August 2023: Nidharssan Selva, summer Intern at IIT Madras
- Dec 2021: GW data analysis sessions (1 month) for winter interns at IIT Madras
- 2020-2021: Preet Baxi, research project student at IIT Madras

Teaching Assistant
- 2019-2024: Physics 101, Physics 102, Foundation of Computational Physics, Electromagnetic Theory — IIT Madras

## Professional Membership

2020-present LIGO Scientific Collaboration (LSC)

2024-present Cosmic Explorer Consortium

## Academic service, Outreach, DEI

**Talks**

- *Cosmic dance of black holes*, **Alumni Talks** (online) at NIT Agartala, IN (June 2022)
- *Black Holes and Gravitational Waves*, online talk for school students of Chennai, IN (October 2021)
- *An introduction to gravitational waves*, **Open Seminar Series** (online), IIT Madras, Chennai, IN (March 2021)

**Academic Service**

- Poster presentation judge: LVK Collaboration Meeting, IUCAA, IN (2026)
- Press release of the special GW event GW230529 – Hindi translations for news articles in India (2024)
- LOC: Second Chennai Symposium on Gravitation and Cosmology, IIT Madras (2022)
- LOC: School on Black Holes and Gravitational Waves, IIT Madras (2022)

**Media & Interviews**

- Community Building Panel – Interview with the LSC DEI committee as a panelist in the LVK Collaboration meeting, Sept 2026
- "Understanding gravitational wave research and its impact on society" – Interview with Science Communication students at Cardiff University (2024)
- Glorious Women of Astronomy – Interview with LIGO India team (2021)

**Publications**

4. *"LAAC Corner: Thesis Writing"*, article in LIGO magazine 2026
3. *"A Farewell"*, science fiction story in Science Reporter, SR Vol. 55(11), Page 48-49 (2018)
2. *"Beyond the Blackhole"*, science fiction story in Anthology – Alternate Reality by Cypher Publications (2015)
1. *"Evolution A Question"*, science fiction story in Science Reporter, SR Vol. 50(12), Page 36-38 (2013)

## Software Projects

1. `quick_bib`: A public python package to assist in management of bib entries and bib files. `https://github.com/divyajyoti09/quick_bib`
2. Management of workstation "powehi" at the Department of Physics, IIT Madras, which is used exclusively by the gravitational wave group led by Dr. Chandra Kant Mishra.

## Research Publications: Short-Author*

*includes works where a short non-alphabetical list of authors (including Divyajyoti) is followed by the alphabetical list from large collaborations
ORCID: 0000-0002-2787-1012, InspireHEP bibliography

12. LIGO A♯: Detector Design and Science Prospects Beyond A+ — *L. Sun, K. Kuns, B.J.J. Slagmolen, P. Fritschel, P. Schmidt, B.T. Lantz, S.S.Y. Chua,* ***Divyajyoti*** *et al.* (2026). arXiv:2608.11673

11. Long-duration GW Searches for Sub-solar NSs and Superkilonovae using CoCoA — *A. Tejera, **Divyajyoti** , A. Corsi, Y. Zenati, R. Coyne, L. Cote.* (2026). arXiv:2608.11387
10. Radio Follow Up of a Sub-threshold GRB in the Sky Localization Area of GW241125 — *N. Gottschlich, A. Corsi, S. B. Cenko, **Divyajyoti** , J. DeLaunay, D. B. Fox, T. O'Dwyer, S. Ronchini.* (2026). arXiv:2606.20885
9. Mapping the star formation peak with the next-generation gravitational wave detectors — ***Divyajyoti**, M. K. Singh, S. Fairhurst, M. Hannam.* (2026). arXiv:2606.05151
8. Biased parameter inference of eccentric, spin-precessing binary black holes — ***Divyajyoti**, I. M. Romero-Shaw, V. Prasad, K. Paul, C. K. Mishra, P. Kumar, A. Maurya, M. Boyle, L. E. Kidder, H. P. Pfeiffer, M. A. Scheel.* Phys. Rev. D 113, 103022 (2026). arXiv:2510.04332
7. Inferring additional physics through unmodelled signal reconstructions — *R. Das, Gayathri V, **Divyajyoti**, S. Jose, I. Bartos, S. Klimenko, C. K. Mishra.* Phys. Rev. D 112, 023011 (2025). arXiv:2412.11749
6. ESIGMAHM: An Eccentric, Spinning inspiral-merger-ringdown waveform model with Higher Modes for the detection and characterization of binary black holes — *K.Paul, A. Maurya, Q. Henry, K. Sharma, P. Satheesh, **Divyajyoti**, P. Kumar, C. K. Mishra.* Phys. Rev. D 111, 084074 (2025). arXiv:2409.13866
5. Blind spots and biases: the dangers of ignoring eccentricity in gravitational-wave signals from binary black holes — ***Divyajyoti**, S. Kumar, S. Tibrewal, I. M. Romero-Shaw, C. K. Mishra.* Phys. Rev. D 109, 043037 (2024). arXiv:2309.16638
4. Effect of double spin-precession and higher harmonics on spin-induced quadrupole moment measurements — ***Divyajyoti**, N. V. Krishnendu, M. Saleem, M. Colleoni, A. Vijaykumar, K. G. Arun, C. K. Mishra.* Phys. Rev. D 109, 023016 (2024). arXiv:2311.05506
3. High accuracy PN and NR comparisons involving higher modes for eccentric BBHs and a dominant mode eccentric IMR model — *A. Chattaraj, T. RoyChowdhury, **Divyajyoti**, C. K. Mishra, A. Gupta.* Phys. Rev. D 106, 124008 (2022). arXiv:2204.02377
2. Improved early-warning estimates of luminosity distance and orbital inclination of compact binary mergers using higher modes of gravitational radiation — *M. K. Singh, **Divyajyoti**, S. J. Kapadia, M. A. Shaikh, P. Ajith.* Mon. Not. Roy. Astron. Soc. 513(3), 3798-3809 (2022). arXiv:2202.05802
1. Detectability of gravitational higher modes in the third-generation era — ***Divyajyoti**, P. Baxi, C. K. Mishra, and K. G. Arun.* Phys. Rev. D 104, 084080 (2021). arXiv:2103.03241

## Research Publications: Large collaboration

The publications listed here had active contribution from me. To see all papers in which I am listed as an author, please visit my InspireHEP bibliography

5. GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog — *The LVK Collaboration (incl. **Divyajyoti**).* (2026). arXiv:2605.27225

4. GWTC-4.0: Tests of General Relativity. II. Parameterized Tests — *The LVK Collaboration (incl. **Divyajyoti**)*. (2026). arXiv:2603.19020
3. GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run — *The LVK Collaboration (incl. **Divyajyoti**)*. Astrophys. J. Lett. 1004 2, L22 (2026). arXiv:2508.18082
2. Tests of general relativity with GWTC-3 — *The LVK Collaboration (incl. **Divyajyoti**)*. Phys. Rev. D 112, 084080 (2025). arXiv:2112.06861
1. GWTC-2.1: Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run — *The LVK Collaboration (incl. **Divyajyoti**)*. Phys. Rev. D 109, 022001 (2024). arXiv:2108.01045